

HEALTH PLUS CARE

DATA PROFILING:

1. Check the structure of tables.

describe Specialists;

2. View all records in each table.

select * from clinics;

3. Count the total number of records in each table.

select count(*) from Members;

4. Find inconsistent entries in a column.

SELECT distinct gender from Members;

5. Find records with null entries in a column.

SELECT * from Members where email is null;

6. Find incorrect email.

SELECT * from Members where email not like '%@%.%';

SELECT *

FROM Members

WHERE email LIKE '%@@%';

7. Find incorrect values in total_amount column of billing table.

SELECT

bill_id,

consultation_charges,

lab_charges,

medicine_charges,

total_amount,

consultation_charges + lab_charges + medicine_charges AS calculated_total

FROM Billing

WHERE total_amount < 0;

8. Find incorrect values in payment_amount column of payments table.

SELECT

payment_id,

bill_id,

payment_amount,

FROM Payments

```
WHERE payment_amount < 0;
SELECT COUNT(*) AS negative_payments
FROM Payments WHERE payment_amount < 0;
```

DATA CLEANING:

a. Standardizing inconsistent entries in gender column of members table.

```
UPDATE Members
set gender=
CASE
WHEN lower(trim(gender)) in ('male','m')
THEN 'Male'
WHEN lower(trim(gender)) in ('female','f')
THEN 'Female'
ELSE gender
END;
```

b. Correcting incorrect email entries from email column of members table.

```
update Members
set email=
CASE
WHEN email like '%gmail' and email not like '%@%'
THEN REPLACE(email,'gmail','@gmail')
END
where email not like '%@%.%';
```

c. Standardizing inconsistent entries in status column of consultations table.

```
UPDATE Consultations
SET status =
CASE
    WHEN LOWER(TRIM(status)) = 'completed' THEN 'Completed'
    WHEN LOWER(TRIM(status)) = 'cancelled' THEN 'Cancelled'
    WHEN LOWER(TRIM(status)) = 'scheduled' THEN 'Scheduled'
    WHEN LOWER(TRIM(status)) = 'no-show' THEN 'No-Show'
    ELSE status
END;
```

d. Modifying consultations table to convert string values to data values in consultation_date column.

```
ALTER TABLE Consultations
MODIFY consultation_date DATE NOT NULL;
```

e. Updating correct values in total_amount column of billing column.

```
UPDATE Billing
SET total_amount =
    consultation_charges
    + lab_charges
    + medicine_charges
WHERE total_amount < 0;
```

f. Updating correct values in payment_amount column of payments table.

```
UPDATE Payments
SET payment_amount = ABS(payment_amount)
WHERE payment_amount < 0;
```

g. Filling null values

```
UPDATE Claims
SET insurance_provider = 'Unknown'
WHERE insurance_provider IS NULL;
```

h. Adding extra column in members table to give email status as valid or invalid.

```
ALTER TABLE Members
ADD COLUMN email_status VARCHAR(10);
UPDATE Members
SET email_status =
CASE
    WHEN email IS NULL THEN 'Invalid'
    WHEN email NOT LIKE '%@%.%' THEN 'Invalid'
    WHEN email LIKE '%@@%' THEN 'Invalid'
    ELSE 'Valid'
END;
```

i. Handling missing values in Claims table

```
UPDATE Claims
SET insurance_provider = 'Unknown'
WHERE insurance_provider IS NULL;
```

DATA ANALYSIS:

1. Which clinics receive the highest consultation activity?

```
SELECT
    c.clinic_id, cl.clinic_name,
    COUNT(c.consultation_id) AS consultation_count
FROM Consultations c
JOIN Clinics cl
    ON c.clinic_id = cl.clinic_id
GROUP BY
    c.clinic_id, cl.clinic_name
ORDER BY consultation_count DESC;
```

	clinic_id	clinic_name	consultation_count
▶	C001	HealthPlus Diagnostic Clinic - Kozhikode	945
	C017	HealthPlus Diagnostic Clinic - Pune	337
	C012	HealthPlus Primary Care Clinic - Mangaluru	317
	C004	HealthPlus Primary Care Clinic - Delhi	304
	C003	HealthPlus Telemedicine Hub Clinic - Hyderabad	303
	C015	HealthPlus Telemedicine Hub Clinic - Hyderabad	300
	C013	HealthPlus Diagnostic Clinic - Kozhikode	298
	C011	HealthPlus Telemedicine Hub Clinic - Pune	297
	C007	HealthPlus Telemedicine Hub Clinic - Coimbatore	296
	C009	HealthPlus Diagnostic Clinic - Hyderabad	296
	C018	HealthPlus Wellness Clinic - Kozhikode	296
	C016	HealthPlus Primary Care Clinic - Madurai	295
	C002	HealthPlus Wellness Clinic - Kozhikode	293

2. Which specialists have the highest consultation workload?

```
SELECT
    s.specialist_id,
    CONCAT(s.first_name, ' ', s.last_name) AS specialist_name,
    s.specialization,
    COUNT(c.consultation_id) AS consultation_count
FROM Specialists s
JOIN Consultations c
    ON s.specialist_id = c.specialist_id
GROUP BY
    s.specialist_id,
    specialist_name,
    s.specialization
ORDER BY consultation_count DESC;
```

	specialist_id	specialist_name	specialization	consultation_count
▶	SP0004	Charu Rao	ENT Specialist	391
	SP0093	Sameer Malhotra	Pulmonologist	70
	SP0080	Rajat Saxena	Physiotherapist	66
	SP0133	Sanya Ahluwalia	Pediatrician	65
	SP0112	Nikhil Menon	Pulmonologist	62
	SP0194	Sanya Bhatia	Cardiologist	59
	SP0123	Anjali Choudhary	Physiotherapist	55
	SP0088	Meghna Choudhary	Dietitian	53
	SP0086	Diya Bhatia	Diabetologist	52
	SP0118	Mansi Desai	Diabetologist	48
	SP0155	Preeti Kapoor	Nephrologist	48
	SP0173	Priyal Tiwari	Dermatologist	48
	SP0055	Kirti Chawla	Dietitian	46

3. Which specializations have the highest activity?

```

SELECT
    s.specialization,
    COUNT(c.consultation_id) AS consultation_count
FROM Specialists s
JOIN Consultations c
    ON s.specialist_id = c.specialist_id
GROUP BY s.specialization
ORDER BY consultation_count DESC;

```

	specialization	consultation_count
▶	ENT Specialist	744
	Dietitian	531
	Nephrologist	477
	Physiotherapist	452
	Psychiatrist	431
	Pulmonologist	428
	Diabetologist	419
	Dentist	407
	Endocrinologist	360
	Cardiologist	356
	Dermatologist	348
	Gynecologist	322
	Pediatrician	305

4. Which members are the most active healthcare users?

```

SELECT
    m.member_id,

```

```

        CONCAT(m.first_name, ' ', m.last_name) AS member_name,
        m.membership_type,
        COUNT(c.consultation_id) AS consultation_count
FROM Members m
JOIN Consultations c
    ON m.member_id = c.member_id
GROUP BY
    m.member_id,
    member_name,
    m.membership_type
ORDER BY consultation_count DESC;

```

	member_id	member_name	membership_type	consultation_count
▶	M001338	Rohit Chawla	Individual	11
	M001362	Rajat Joshi	Family	9
	M001441	Charu Verma	Family	9
	M000028	Dev Kapoor	Family	8
	M000112	Ojasvi Ahluwalia	Individual	8
	M000555	Charu Saxena	Family	8
	M001917	Tanvi Khanna	Individual	8
	M002095	Mansi Choudhary	Family	8
	M000035	Shaurya Sharma	Corporate	7
	M000126	Rahul Mehta	Individual	7
	M000140	Arvind Chawla	Individual	7
	M000191	Riya Chawla	Family	7
	M000476	Priyal Patel	Individual	7
	M000494	Kavita Bhatia	Individual	7

5. How do consultation modes compare?

```

SELECT
    consultation_mode,
    COUNT(*) AS consultation_count,
    ROUND(
        COUNT(*) * 100.0 / (SELECT COUNT(*) FROM Consultations),
        2
    ) AS percentage
FROM Consultations
GROUP BY consultation_mode
ORDER BY consultation_count DESC;

```

	consultation_mode	consultation_count	percentage
▶	In-Clinic	3522	58.70
	Telemedicine	2478	41.30

6. Which consultation statuses require attention?

```

SELECT
    status,
    COUNT(*) AS consultation_count,

```

```

ROUND(
    COUNT(*) * 100.0 / (SELECT COUNT(*) FROM Consultations),
    2
) AS percentage
FROM Consultations
GROUP BY status
ORDER BY consultation_count DESC;

```

	status	consultation_count	percentage
▶	Completed	4400	73.33
	Scheduled	594	9.90
	Cancelled	530	8.83
	No-Show	476	7.93

7. How many telemedicine sessions are completed, cancelled, or in other statuses?

```

SELECT
    session_status,
    COUNT(*) AS session_count,
    ROUND(
        COUNT(*) * 100.0 / (SELECT COUNT(*) FROM Telemedicine_Sessions),
        2
    ) AS percentage
FROM Telemedicine_Sessions
GROUP BY session_status
ORDER BY session_count DESC;

```

	session_status	session_count	percentage
▶	Completed	1465	66.59
	Dropped	735	33.41

8. Which telemedicine platforms and connection-quality categories are most common?

```

SELECT
    platform,
    connection_quality,
    COUNT(*) AS session_count
FROM Telemedicine_Sessions
GROUP BY
    platform,
    connection_quality
ORDER BY session_count DESC;

```

	platform	connection_quality	session_count
▶	App	Good	561
	Phone Call	Good	516
	Web	Good	490
	App	Average	164
	Phone Call	Average	145
	Web	Average	136
	Phone Call	Poor	70
	App	Poor	67
	Web	Poor	51

9. What is the average telemedicine session duration?

```
SELECT
  ROUND(
    AVG(
      TIMESTAMPDIFF(
        MINUTE,
        session_start_time,
        session_end_time
      )
    ),
    2
  ) AS avg_session_duration_minutes
FROM Telemedicine_Sessions
WHERE session_start_time IS NOT NULL
AND session_end_time IS NOT NULL;
```

	avg_session_duration_minutes
▶	21.43

10. Which chronic conditions have the highest program enrollment?

```
SELECT
  condition_name,
  COUNT(*) AS program_enrollment
FROM Chronic_Care_Programs
GROUP BY condition_name
ORDER BY program_enrollment DESC;
```

	condition_name	program_enrollment
▶	Hypertension	120
	Type 2 Diabetes	108
	Asthma	107
	Obesity Management	102
	Arthritis	100
	Thyroid Disorder	96
	Chronic Kidney Disease	92
	Chronic Back Pain	90
	Cardiac Risk Management	85

11. Which specialists manage the most chronic-care programs?

```
SELECT
    s.specialist_id,
    CONCAT(s.first_name, ' ', s.last_name) AS specialist_name,
    s.specialization,
    COUNT(ccp.program_id) AS chronic_care_programs
FROM Specialists s
JOIN Chronic_Care_Programs ccp
    ON s.specialist_id = ccp.specialist_id
GROUP BY
    s.specialist_id,
    specialist_name,
    s.specialization
ORDER BY chronic_care_programs DESC;
```

	specialist_id	specialist_name	specialization	chronic_care_programs
▶	SP0028	Kabir Ahluwalia	Psychiatrist	10
	SP0195	Manish Menon	Orthopedic	10
	SP0056	Nikhil Chawla	Physiotherapist	9
	SP0103	Karan Rao	Physiotherapist	9
	SP0105	Puneet Iyer	Orthopedic	9
	SP0124	Diya Tiwari	Pulmonologist	9
	SP0217	Rajat Malhotra	Nephrologist	9
	SP0062	Kavita Chopra	Gynecologist	8
	SP0166	Nidhi Malhotra	Orthopedic	8
	SP0009	Garima Ahluwalia	Dentist	7
	SP0031	Arvind Malhotra	Physiotherapist	7
	SP0045	Gaurav Kulkarni	Psychiatrist	7
	SP0080	Rajat Saxena	Physiotherapist	7
	SP0001	Pooja Singh	Physiotherapist	7

12. Which health packages have the highest subscription volume?

```
SELECT
    hp.package_id,
    hp.package_name,
    hp.package_type,
    COUNT(ps.subscription_id) AS subscription_count
FROM Health_Packages hp
JOIN Package_Subscriptions ps
    ON hp.package_id = ps.package_id
GROUP BY
    hp.package_id,
    hp.package_name,
    hp.package_type
ORDER BY subscription_count DESC;
```

	package_id	package_name	package_type	subscription_count
▶	PK002	Full Body Checkup Premium v2	Premium	92
	PK023	Cardiac Risk Package v23	Cardiac	89
	PK006	Full Body Checkup Premium v6	Premium	86
	PK019	Senior Citizen Care Package v19	Senior Citizen	83
	PK025	Senior Citizen Care Package v25	Senior Citizen	82
	PK003	Senior Citizen Care Package v3	Senior Citizen	81
	PK008	Diabetes Care Package v8	Diabetes Care	81
	PK009	Basic Health Checkup v9	Basic	81
	PK017	Basic Health Checkup v17	Basic	80
	PK005	Cardiac Risk Package v5	Cardiac	80
	PK029	Women's Wellness Package v29	Wellness	80
	PK028	Women's Wellness Package v28	Wellness	80
	PK022	Basic Health Checkup v22	Basic	79
	PK031	Executive Health Checkup v31	Executive	78

13. Which subscriptions are approaching or have passed expiry?

SELECT

ps.subscription_id,

m.member_id,

hp.package_name,

ps.subscription_date,

ps.expiry_date,

ps.payment_status,

CASE

WHEN ps.expiry_date < CURDATE() THEN 'Expired'

WHEN ps.expiry_date <= DATE_ADD(CURDATE(), INTERVAL 30 DAY)

THEN 'Expiring within 30 days'

ELSE 'Active'

END AS subscription_status

FROM Package_Subscriptions ps

JOIN Members m

ON ps.member_id = m.member_id

JOIN Health_Packages hp

ON ps.package_id = hp.package_id

ORDER BY ps.expiry_date;

	subscription_id	member_id	package_name	subscription_date	expiry_date	payment_status	subscription_status
▶	PS00297	M001791	Cardiac Risk Package v37	2024-01-01	2024-06-29	Refunded	Expired
	PS00595	M000165	Diabetes Care Package v8	2024-01-03	2024-07-01	Paid	Expired
	PS01654	M001906	Cardiac Risk Package v5	2024-01-03	2024-07-01	Paid	Expired
	PS00254	M000637	Cardiac Risk Package v37	2024-01-06	2024-07-04	Paid	Expired
	PS00720	M002305	Cardiac Risk Package v37	2024-01-06	2024-07-04	Paid	Expired
	PS01832	M000023	Diabetes Care Package v8	2024-01-08	2024-07-06	Paid	Expired
	PS00644	M001733	Cardiac Risk Package v5	2024-01-09	2024-07-07	Paid	Expired
	PS01581	M000176	Diabetes Care Package v8	2024-01-09	2024-07-07	Paid	Expired
	PS01872	M000107	Cardiac Risk Package v37	2024-01-09	2024-07-07	Refunded	Expired
	PS02673	M000105	Cardiac Risk Package v13	2024-01-09	2024-07-07	Paid	Expired
	PS01431	M000096	Cardiac Risk Package v5	2024-01-10	2024-07-08	Paid	Expired
	PS02627	M001684	Cardiac Risk Package v12	2024-01-10	2024-07-08	Paid	Expired
	PS01177	M000504	Cardiac Risk Package v16	2024-01-11	2024-07-09	Paid	Expired
	PS00866	M000932	Diabetes Care Package v15	2024-01-12	2024-07-10	Paid	Expired

14. Which corporates contribute the most enrolled members?

```
SELECT
    c.corporate_id,
    c.company_name,
    c.industry,
    c.employee_count,
    COUNT(cm.corporate_member_id) AS enrolled_members
FROM Corporates c
JOIN Corporate_Members cm
    ON c.corporate_id = cm.corporate_id
GROUP BY
    c.corporate_id,
    c.company_name,
    c.industry,
    c.employee_count
ORDER BY enrolled_members DESC;
```

	corporate_id	company_name	industry	employee_count	enrolled_members
▶	CO036	GlobalSoft Pvt Ltd	Education	1000	18
	CO044	DeltaSolutions Pvt Ltd	Consulting	4551	18
	CO042	ZenSolutions Pvt Ltd	Consulting	4991	18
	CO019	TechSystems Pvt Ltd	Telecommunications	2549	16
	CO058	NovaSolutions Pvt Ltd	Telecommunications	581	16
	CO027	SkyIndustries Pvt Ltd	Retail	2418	15
	CO007	SkyIndustries Pvt Ltd	E-commerce	2407	13
	CO010	NovaSolutions Pvt Ltd	Retail	4279	13
	CO026	SkyIndustries Pvt Ltd	Education	2950	13
	CO017	DeltaSystems Pvt Ltd	Pharmaceuticals	1174	13
	CO020	ApexLogistics Pvt Ltd	Telecommunications	1894	13
	CO008	TechSystems Pvt Ltd	Banking & Finance	3811	12
	CO002	ApexSolutions Pvt Ltd	Banking & Finance	1818	12
	CO031	ZenIndustries Pvt Ltd	Manufacturing	3256	12

15. Which industries have the greatest corporate healthcare participation?

```
SELECT
    c.industry,
    COUNT(cm.corporate_member_id) AS enrolled_members
FROM Corporates c
JOIN Corporate_Members cm
    ON c.corporate_id = cm.corporate_id
GROUP BY c.industry
ORDER BY enrolled_members DESC;
```

	industry	enrolled_members
►	Consulting	126
	Education	108
	Telecommunications	73
	Pharmaceuticals	63
	Retail	60
	E-commerce	56
	Manufacturing	54
	Banking & Finance	39
	Information Technology	30
	Logistics	6

16. Which medicines are prescribed most frequently?

```

SELECT
    medicine_name,
    COUNT(*) AS prescription_count
FROM Prescriptions
GROUP BY medicine_name
ORDER BY prescription_count DESC;

```

	medicine_name	prescription_count
►	Paracetamol 650mg	554
	Vitamin D3	528
	Cetirizine 10mg	505
	Amlodipine 5mg	501
	Sertraline 50mg	497
	Metformin 500mg	496
	Montelukast 10mg	489
	Losartan 50mg	481
	Atorvastatin 10mg	475
	Pantoprazole 40mg	474

17. Which specialists generate the highest prescription volume?

```

SELECT
    s.specialist_id,
    CONCAT(s.first_name, ' ', s.last_name) AS specialist_name,
    s.specialization,
    COUNT(p.prescription_id) AS prescription_count
FROM Specialists s
JOIN Prescriptions p
    ON s.specialist_id = p.specialist_id
GROUP BY

```

```

s.specialist_id,
specialist_name,
s.specialization
ORDER BY prescription_count DESC;

```

	specialist_id	specialist_name	specialization	prescription_count
▶	SP0004	Charu Rao	ENT Specialist	320
	SP0080	Rajat Saxena	Physiotherapist	59
	SP0093	Sameer Malhotra	Pulmonologist	57
	SP0112	Nikhil Menon	Pulmonologist	53
	SP0116	Sneha Bhatia	Psychiatrist	52
	SP0133	Sanya Ahluwalia	Pediatrician	52
	SP0055	Kirti Chawla	Dietitian	47
	SP0109	Falguni Mishra	Nephrologist	46
	SP0172	Mansi Gupta	Dietitian	45
	SP0155	Preeti Kapoor	Nephrologist	44
	SP0194	Sanya Bhatia	Cardiologist	44
	SP0071	Mansi Nair	Dermatologist	43
	SP0220	Gaurav Kulkarni	General Physic...	43

18. Which lab tests generate the highest total cost?

```

SELECT
    test_name,
    COUNT(*) AS test_count,
    SUM(test_cost) AS total_test_cost
FROM Lab_Tests
GROUP BY test_name
ORDER BY total_test_cost DESC;

```

	test_name	test_count	total_test_cost
▶	Liver Function Test	340	836750.77
	Kidney Function Test	342	832035.98
	Complete Blood Count	352	813212.53
	Chest X-Ray	324	786687.56
	Urine Routine	324	754800.96
	ECG	320	751054.54
	HbA1c	300	718936.38
	Vitamin D Test	303	701759.72
	Lipid Profile	298	691751.60
	Thyroid Profile	307	689588.49
	Vitamin B12 Test	290	648863.62

19. Which clinics have the highest laboratory workload?

```
SELECT
    c.clinic_id,
    c.clinic_name,
    COUNT(lt.lab_test_id) AS lab_test_count
FROM Clinics c
JOIN Lab_Tests lt
    ON c.clinic_id = lt.clinic_id
GROUP BY
    c.clinic_id,
    c.clinic_name
ORDER BY lab_test_count DESC;
```

	clinic_id	clinic_name	lab_test_count
▶	C001	HealthPlus Diagnostic Clinic - Kozhikode	515
	C014	HealthPlus Wellness Clinic - Coimbatore	210
	C004	HealthPlus Primary Care Clinic - Delhi	190
	C006	HealthPlus Wellness Clinic - Coimbatore	189
	C009	HealthPlus Diagnostic Clinic - Hyderabad	187
	C010	HealthPlus Wellness Clinic - Mysuru	185
	C011	HealthPlus Telemedicine Hub Clinic - Pune	185
	C007	HealthPlus Telemedicine Hub Clinic - Coimbatore	183
	C017	HealthPlus Diagnostic Clinic - Pune	181
	C008	HealthPlus Primary Care Clinic - Vijayawada	177
	C005	HealthPlus Diagnostic Clinic - Pune	176
	C015	HealthPlus Telemedicine Hub Clinic - Hyderabad	169
	C012	HealthPlus Primary Care Clinic - Mangaluru	167
	C013	HealthPlus Primary Care Clinic - Mangaluru	166

20. Which insurance providers have the highest claim amount?

```
SELECT
    insurance_provider,
    COUNT(claim_id) AS claim_count,
    SUM(claim_amount) AS total_claim_amount
FROM Claims
GROUP BY insurance_provider
ORDER BY total_claim_amount DESC;
```

	insurance_provider	claim_count	total_claim_amount
▶	Niva Bupa Health Insurance	315	2516309.27
	Care Health Insurance	286	2423825.16
	ICICI Lombard General Insurance	284	2235372.06
	Bajaj Allianz General Insurance	280	2186622.19
	Star Health Insurance	277	2183205.07
	HDFC ERGO Health Insurance	286	2113584.19
	Unknown	72	583134.32

21. What is the distribution of claim statuses?

```
SELECT
    claim_status,
    COUNT(*) AS claim_count,
    ROUND(
        COUNT(*) * 100.0 / (SELECT COUNT(*) FROM Claims),
        2
    ) AS percentage
FROM Claims
GROUP BY claim_status
ORDER BY claim_count DESC;
```

	claim_status	claim_count	percentage
▶	Approved	616	34.22
	Settled	547	30.39
	Pending	471	26.17
	Rejected	166	9.22

22. What is the total billed amount and how is it split among consultation, laboratory, and medicine charges?

```
SELECT
    ROUND(SUM(consultation_charges), 2) AS total_consultation_charges,
    ROUND(SUM(lab_charges), 2) AS total_lab_charges,
    ROUND(SUM(medicine_charges), 2) AS total_medicine_charges,
    ROUND(SUM(total_amount), 2) AS total_billed_amount
FROM Billing;
```

	total_consultation_charges	total_lab_charges	total_medicine_charges	total_billed_amount
▶	4333765.76	10404310.21	4475161.28	19213237.25

23. What is the total payment amount by payment status and payment mode?

```
SELECT
    payment_status,
    payment_mode,
    COUNT(*) AS payment_count,
    ROUND(SUM(payment_amount), 2) AS total_payment_amount
FROM Payments
GROUP BY
    payment_status,
    payment_mode
ORDER BY total_payment_amount DESC;
```

	payment_status	payment_mode	payment_count	total_payment_amount
▶	Success	Cash	1156	3686059.66
	Success	Net Banking	891	2849303.03
	Success	Credit Card	881	2797275.16
	Success	UPI	851	2756935.67
	Success	Wallet	837	2681094.32
	Success	Debit Card	806	2586811.74
	Failed	Cash	81	271304.72
	Failed	Credit Card	77	234621.10
	Failed	Wallet	68	226023.38
	Failed	Net Banking	66	217394.72
	Failed	UPI	72	214397.67
	Failed	Debit Card	62	213762.74
	Refunded	Cash	40	138407.36
	Refunded	Credit Card	35	76771.62

24. What is the validated collection gap?

```

SELECT
    ROUND((SELECT SUM(total_amount) FROM Billing), 2) AS total_billed_amount,
    ROUND((SELECT SUM(payment_amount) FROM Payments), 2) AS
total_payment_amount,
    ROUND(
        (SELECT SUM(total_amount) FROM Billing)
        -
        (SELECT SUM(payment_amount) FROM Payments),
        2
    ) AS collection_gap;

```

	total_billed_amount	total_payment_amount	collection_gap
▶	19213237.25	19213237.25	0.00

25. Which clinics or specialists receive the highest and lowest average feedback ratings?

```

SELECT
    s.specialist_id,
    CONCAT(s.first_name, ' ', s.last_name) AS specialist_name,
    s.specialization,
    COUNT(f.feedback_id) AS feedback_count,
    ROUND(AVG(f.rating), 2) AS average_rating
FROM Specialists s
JOIN Consultations c
    ON s.specialist_id = c.specialist_id
JOIN Feedback f
    ON c.consultation_id = f.consultation_id

```

GROUP BY

s.specialist_id,
specialist_name,
s.specialization

ORDER BY average_rating DESC;

	specialist_id	specialist_name	specialization	feedback_count	average_rating
▶	SP0175	Priyal Chawla	Diabetologist	3	5.00
	SP0001	Nidhi Mishra	Nephrologist	4	5.00
	SP0128	Rahul Bhatia	Dietitian	1	5.00
	SP0105	Puneet Iyer	Orthopedic	5	4.80
	SP0102	Aditi Chopra	Diabetologist	9	4.67
	SP0051	Diya Mishra	Dentist	12	4.58
	SP0200	Nikhil Verma	Dietitian	7	4.57
	SP0164	Shaurya Menon	Dentist	9	4.56
	SP0136	Ankit Malhotra	Orthopedic	8	4.50
	SP0209	Charu Menon	Psychiatrist	8	4.50
	SP0212	Isha Iyer	Endocrinologist	4	4.50
	SP0196	Mansi Iyer	Dermatologist	2	4.50
	SP0130	Nisha Shah	Nephrologist	2	4.50
	SP0027	Vivek Choudhary	Psychiatrist	4	4.50

26. Are there members with consultations but no feedback?

SELECT

m.member_id,
CONCAT(m.first_name, ' ', m.last_name) AS member_name,
COUNT(DISTINCT c.consultation_id) AS consultation_count

FROM Members m

JOIN Consultations c

ON m.member_id = c.member_id

LEFT JOIN Feedback f

ON c.consultation_id = f.consultation_id

WHERE f.feedback_id IS NULL

GROUP BY

m.member_id,
member_name

ORDER BY consultation_count DESC;

	member_id	member_name	consultation_count
►	M001338	Rohit Chawla	11
	M001441	Charu Verma	8
	M000555	Charu Saxena	7
	M001362	Rajat Joshi	7
	M000126	Rahul Mehta	6
	M000526	Nikhil Tiwari	6
	M000587	Ankit Kapoor	6
	M000607	Mansi Gupta	6
	M001021	Aryan Shah	6
	M001556	Meghna Kulkarni	6
	M002041	Amit Rao	6
	M002150	Karan Bansal	6
	M002438	Aditi Shah	6
	M000000	Sanjay Patel	5

27. Are there registered members with no consultations?

```

SELECT
    m.member_id,
    CONCAT(m.first_name, ' ', m.last_name) AS member_name,
    m.membership_type
FROM Members m
LEFT JOIN Consultations c
    ON m.member_id = c.member_id
WHERE c.consultation_id IS NULL
ORDER BY m.member_id;

```

	member_id	member_name	membership_type
►	M000032	Sanjay Patel	Family
	M000048	Rohit Bhatia	Corporate
	M000069	Pooja Agarwal	Family
	M000082	Pooja Menon	Family
	M000098	Siddharth Chawla	Individual
	M000105	Rajat Desai	Individual
	M000107	Swati Chawla	Individual
	M000108	Diya Desai	Corporate
	M000114	Kirti Tiwari	Family
	M000117	Preeti Kapoor	Individual
	M000138	Rohan Bhatia	Family
	M000145	Riya Bansal	Individual
	M000147	Kirti Menon	Corporate
	M000161	Saloni Menon	Family

28. Which business areas show high activity but weak financial or experience indicators?

```
SELECT
    cl.clinic_id,
    cl.clinic_name,
    COUNT(DISTINCT c.consultation_id) AS consultation_count,
    ROUND(AVG(f.rating), 2) AS average_rating,
    ROUND(SUM(b.total_amount), 2) AS total_billed_amount
FROM Clinics cl
LEFT JOIN Consultations c
    ON cl.clinic_id = c.clinic_id
LEFT JOIN Feedback f
    ON c.consultation_id = f.consultation_id
LEFT JOIN Billing b
    ON c.consultation_id = b.consultation_id
GROUP BY
    cl.clinic_id,
    cl.clinic_name
ORDER BY consultation_count DESC;
```

	clinic_id	clinic_name	consultation_count	average_rating	total_billed_amount
►	C001	HealthPlus Diagnostic Clinic - Kozhikode	945	3.81	2893456.00
	C017	HealthPlus Diagnostic Clinic - Pune	337	3.76	950024.91
	C012	HealthPlus Primary Care Clinic - Mangaluru	317	3.81	929310.47
	C004	HealthPlus Primary Care Clinic - Delhi	304	3.99	835513.72
	C003	HealthPlus Telemedicine Hub Clinic - Hyderabad	303	3.71	927049.01
	C015	HealthPlus Telemedicine Hub Clinic - Hyderabad	300	3.93	834481.18
	C013	HealthPlus Diagnostic Clinic - Kozhikode	298	3.71	904998.98
	C011	HealthPlus Telemedicine Hub Clinic - Pune	297	3.68	966543.63
	C007	HealthPlus Telemedicine Hub Clinic - Coimbatore	296	3.88	940920.99
	C009	HealthPlus Diagnostic Clinic - Hyderabad	296	3.72	862779.53
	C018	HealthPlus Wellness Clinic - Kozhikode	296	3.82	863901.87
	C016	HealthPlus Primary Care Clinic - Madurai	295	3.81	952716.80
	C002	HealthPlus Wellness Clinic - Kozhikode	293	3.59	822378.17
	C008	HealthPlus Primary Care Clinic - Viravawada	290	3.90	780062.18

29. Which reasons for visit occur most frequently?

```
SELECT
    reason_for_visit,
    COUNT(*) AS consultation_count
FROM Consultations
GROUP BY reason_for_visit
ORDER BY consultation_count DESC;
```

	reason_for_visit	consultation_count
►	Diet consultation	525
	Diabetes monitoring	525
	Mental health consultation	523
	Skin allergy	513
	Child health checkup	509
	Routine consultation	499
	Blood pressure check	497
	Joint pain	496
	Annual health review	491
	Fever and cold	486
	Pregnancy consultation	473
	Follow-up visit	463